

Independence
Conditional Independence
Chain rule of probability

Independence

- **Independent random variables** – Two random variables X and Y are said to be *statistically independent* if and only if

$$p(x, y) = p(x)p(y)$$

- More precisely, X and Y are independent iff

$$p(X = x, Y = y) = p(X = x)p(Y = y) \quad \forall x \in X \quad y \in Y$$

- Examples

- Independent – X : Throw of a dice, Y : Toss of a coin
 - Not independent – X : Height, Y : Weight
- Independence is equivalent to saying

$$p(y/x) = p(y) \text{ OR } p(x/y) = p(x)$$

- Can be seen from product rule $p(x, y) = p(y|x)p(x) = p(x)p(y)$
 $\Rightarrow p(y|x) = p(y)$

- Independence is denoted by $x \perp y$

Conditional Independence

- Two random variables X and Y are said to be *independent given Z* if and only if

$$p(x, y | z) = p(x|z)p(y|z)$$

- More precisely, X and Y are independent given Z iff

$$p(X = x, Y = y | Z = z) = p(X = x | Z = z)p(Y = y | Z = z) \quad \forall x \in X, y \in Y, z \in Z$$

Examples

- Ind & Cond Ind* – X : Throw of a dice, Y : Toss of a coin, Z : Card from deck
 - Not Ind BUT Cond Ind* – X : Height, Y : Vocabulary, Z : Age
 - Ind BUT Cond Not Ind* – X : Dice Throw 1, Y : Dice Throw 2, Z : Sum of dice
- Denoted by $x \perp y | z$

$$p(x, y) = p(x)p(y)$$
$$p(x, y | z) = p(x)p(y)$$
$$p(x | z) = p(x)$$
$$p(y | z) = p(y)$$

Not independent

Conditional Probability – Chain rule

- Recall Product Rule : $p(x, y) = p(x|y)p(y)$

Consider $p(x, y, z) = p(x, a)$ where a is the event (y, z)

$$\begin{aligned}\Rightarrow p(x, y, z) &= p(x, a) = \overset{\text{Product Rule}}{p(x|a)p(a)} \\ &= p(x|a)p(y, z) \\ &= p(x|a)p(y|z)p(z) \\ &= p(x|y, z)p(y|z)p(z) \\ \overset{p(x, y, z)}{=} &= p(z)p(y|z)p(x|y, z)\end{aligned}$$

- In general,

- $P(x^{(1)}, x^{(2)}, \dots, x^{(n)}) = P(x^{(1)})P(x^{(2)}|x^{(1)}) \dots P(x^{(n)}|x^{(1)}, \dots, x^{(n-1)})$

i.e.

$$P(x^{(1)}, x^{(2)}, \dots, x^{(n)}) = P(x^{(1)}) \overset{\text{Product}}{\prod_{i=2}^n} P(x^{(i)}|x^{(1)}, \dots, x^{(i-1)})$$

Chain rule of (conditional) probability